Subject: Module 5 Assignment - Order Tracking of Mobius Propeller Data

Date: May 5, 2025 (Submitted)

Order Tracking of Mobius Propeller Data

The Matlab code for this problem is listed in Appendix 1.

Spectrogram for Channel 1

Figure 1 shows the spectrogram for the Channel 1 pressure recording at an angle of +23°.

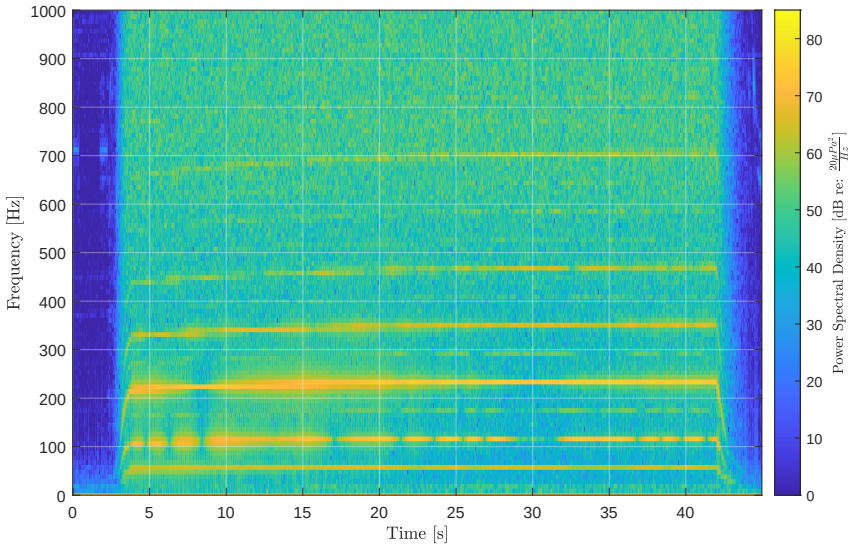

Figure 1: Spectrogram for the sound pressure recording at +23°.

The bottom trace in the spectrogram, which corresponds to shaft rotation, is about 58.6 cycles per second.

Shaft Rotation Speed

Figure 2 shows the estimated rotation speed of the shaft, calculated on a per revolution basis.

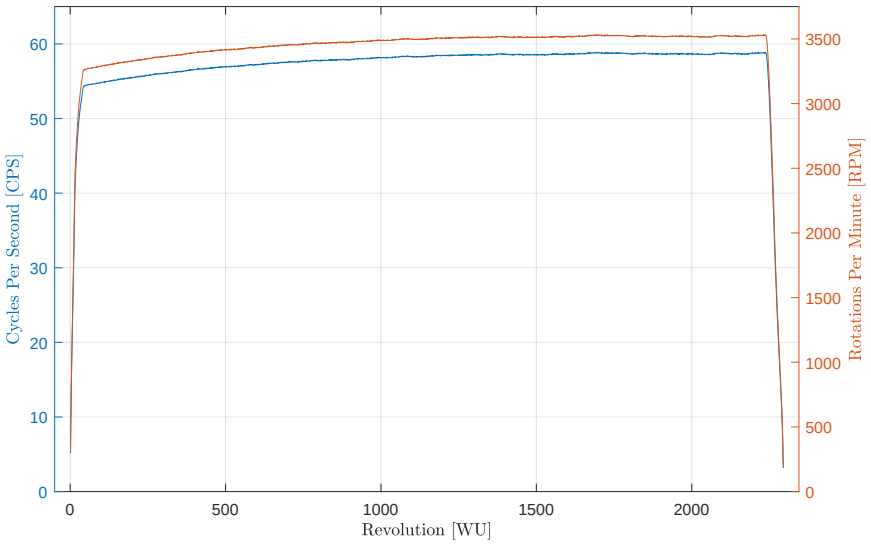

Figure 2: Estimation of shaft rotation speed per revolution.

The maximum shaft rotation speed is approximately 58.9 cycles per second, or 3,532 RPM.

Second-order Tracking for Channel 1

Figure 3 shows the second-order tracking profile for the Channel 1 pressure recording at +23°.

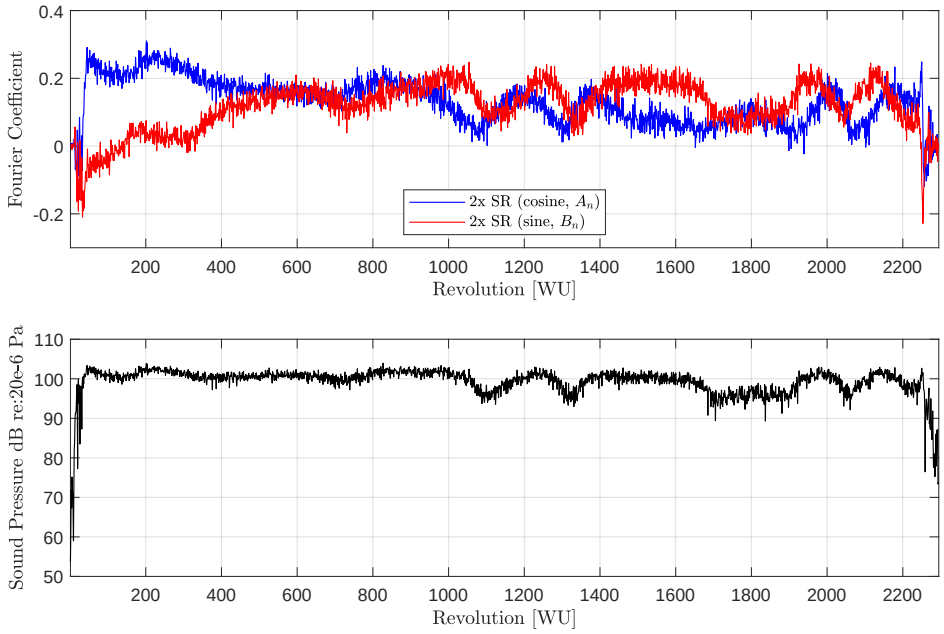

Figure 3: Second-order tracking for Channel 1.

Second-order Tracking Across Channels

Figure 4 shows the second-order tracking profiles for all of the pressure recording channels.

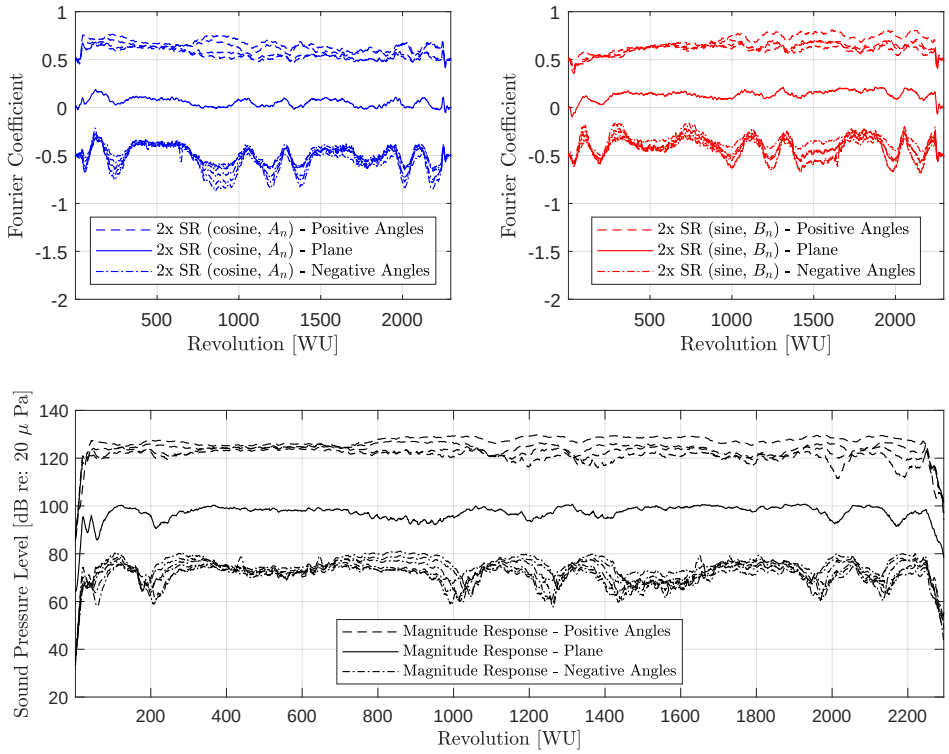

Figure 4: Second-order tracking across channels. A 0.5 offset was used to visually separate the Fourier coefficient profiles. A 20 dB offset was used to separate the sound pressure level profiles.

To visualize the tracking behaviour, the second-order tracking profiles have been segregated into three groups: positive angle channels (1 to 4), a 0° channel (in-plane with the propeller), and negative angle channels (6 to 11). For the Fourier coefficients, an offset of 0.5 was added to the positive angle channels, no offset for the 0° channel, and an offset of -0.5 was added to the negative angle channels. For the sound pressure levels, an offset of 20 dB was used to offset the channel groups.

Fourth-order Tracking Across Channels

Figure 5 shows the fourth-order tracking profiles for all of the pressure recording channels.

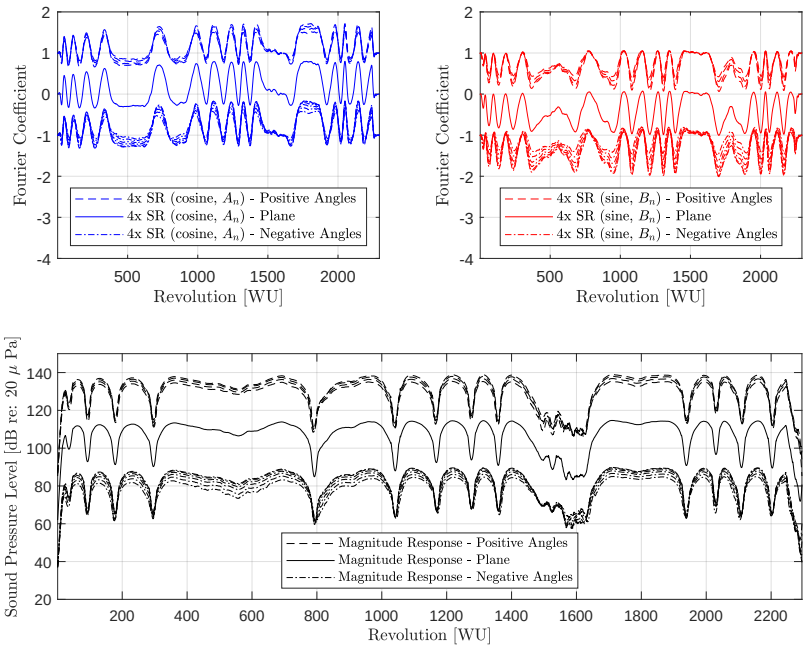

Figure 5: Fourth-order tracking across channels.

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1 Appendix - Matlab Code

```
1
2
3
4   %% Synopsis
5
6   % Problem 5 - Order Tracking of Mobius Propeller Data
7
8   % See Lecture 26
9   %    "D:\15 Downloads\00 GitHub\ACS_547\35 Lectures\26 Monday, April 21, 2025\Lecture 26 - Shafting and
        bearings.pptx"
10
11
12
13  %% Environment
14
15  close all; clear; clc;
16  % restoredefaultpath;
17
18  addpath( genpath( './00 Support' ), '-begin' );
19
20  % set( groot, 'DefaultFigurePosition', [ 230 730  3*750  500 ] );  % x, y, width, height
21
22  set( 0, 'DefaultFigurePaperPositionMode', 'manual' );
23  set( 0, 'DefaultFigureWindowStyle', 'normal' );
24  set( 0, 'DefaultLineLineWidth', 0.6 );
25  set( 0, 'DefaultTextInterpreter', 'Latex' );
26
27  format ShortG;
28
29  color_map = slanCM( 'ColorBlind', 4 );
30
31  pause( 1 );
32
33
34
35  %% Load Data
36
37  load( 'mobius_prop_data.mat' );  % Variable(s):  fs;  mic_angles_degrees;  p;  trigger
38
39  % fs - Sample rate of 80 kHz (scalar).
40  % mic_angles_degrees - Microphone angles relative to horizontal plane (11-by-1;  23, 18, 6, 0, -6, -12,
        -18, -23, -28, and -32).
41  % p - Recorded pressures for the corresponding microphone angles (3,600,000-by-11).
42  % trigger - Shaft rotation synchronization signal (3,600,000-by-11).
43
44
45
46  %% Visualize Experimental Data
47
48  net_time = size( p, 1 ) / fs;  % 45 seconds
49
50  time_indices = ( 0:1:( numel( trigger ) - 1 ) ) ./ fs;
51      time_indices = time_indices(:);
52
53
54  figure( 'Name', 'Trigger signal and Channel 1 Pressure Data' ); ...
55
56      h1 = subplot( 2, 1, 1 ); ...
57          plot( time_indices, trigger, 'Color', color_map( 1, : ) );  grid on;
58              xlabel( 'Time [s]' );  ylabel( 'Voltage [V]' );  title( 'Shaft Trigger Signal' );
59              ylim( [ -0.5 4 ] );
60      h2 = subplot( 2, 1, 2 ); ...
61          plot( time_indices, p( :, 1 ), 'Color', color_map( 2, : ) );  grid on;
62              xlabel( 'Time [s]' );  ylabel( 'Pressure [Pa]' );  title( 'Recorded Pressure at 23$^\circ$
                    Elevation' );
63              ylim( [ -5 5.5 ] );
64
65      linkaxes( [ h1 h2 ], 'x' );
66          xlim( [ -5 50 ] );
67
68
69
70  %% Spectrogram
71
72  signal = p(:, 1);
73
74  frame_length = 8192;
75      frame_hop = floor( 0.20 * frame_length );
76
77  a_spectrogram = spectrogram_September_26_2023( signal, frame_length, frame_hop, fs, 0 );
78
79  figure( 'Name', 'Spectrogram of Channel 1 Pressure Signal' ); ...
80
81      imagesc( a_spectrogram.time_indices, a_spectrogram.Sxx.frequencies, 20*log10( sqrt(a_spectrogram.
            Sxx.spectrum)./20e-6 ), [ 0 85 ] );
82          labelColorbar( 'Power Spectral Density [dB re: $\frac{20 \mu Pa^2}{Hz}$]' );  grid on;
83          colormap parula;  % Option:  Turbo
84          set( gca, 'GridColor', 'w', 'GridAlpha', 0.4 );
85      xlabel( 'Time [s]' );  ylabel( 'Frequency [Hz]' );
```

```
86      set( gca, 'ydir', 'normal' );
87      ylim( [ 0 1e3] );
88
89
90
91  %% Locate the Leading Edges of the Trigger Data
92
93  % Smooth data to reduce noise;  supports finding the leading eges.
94  smoothed_trigger_data = [ 0;  diff( smooth( trigger, 'moving', 5 ) ) ];
95
96  % Threshold smoothed data to located leading edges.
97  threshold = 0.5;
98      edge_indices = find( threshold  <  smoothed_trigger_data );
99          leading_edges = edge_indices( 1:2:end );
100
101 temp = diff( leading_edges );
102     [ indices_to_remove, indices ] = find( temp == 2 );
103         clear temp;
104
105 temp = leading_edges;
106     temp( indices_to_remove ) = [ ];
107
108 leading_edges = temp;
109     clear temp;
110
111 time_indices_leading_edge = leading_edges ./ fs;
112
113
114 % figure( 'Name', 'Revolution Segmentation' ); ...
115 %
116 %     plot( trigger( 1:1:end ), 'Color', 'k' );  hold on;
117 %
118 %     for index = 1:1:numel( leading_edges )
119 %         line( [ leading_edges( index ) leading_edges( index ) ], [ 0 4 ], 'Color', 'b' );  grid on;
120 %     end
121 %
122 %     axis( [ 1 numel( trigger )  -0.5  4.5 ] );
123
124
125
126 %% Segment Microphone Recording Channels
127
128 % Channel 1:  23 degrees
129 % Channel 2:  18 degrees
130 % Channel 3:  12 degrees
131 % Channel 4:  6 degrees
132 % Channel 5:  0 degrees
133 % Channel 6:  -6 degrees
134 % Channel 7:  -12 degrees
135 % Channel 8:  -18 degrees
136 % Channel 9:  -23 degrees
137 % Channel 10:  -28 degrees
138 % Channal 11:  -32 degrees
139
140
141 start_indices = 1:1:( numel( leading_edges ) - 1 );  end_indices = 2:1:numel( leading_edges );
142
143 frame_set_indices = [ leading_edges( start_indices(:) )  leading_edges( end_indices(:) ) ];
144     number_of_revolutions = size( frame_set_indices, 1 );
145
146
147 for channel_index = 1:1:size( p, 2 )
148
149     for frame_index = 1:1:size( frame_set_indices, 1 )
150
151         channel_segments{ frame_index, channel_index } = p( frame_set_indices( frame_index, 1 ):1:
                frame_set_indices( frame_index, 2 ), channel_index );
152
153         slope_theta = (2*pi) ./ size( channel_segments{ frame_index, channel_index }, 1 );
154             theta{ frame_index, channel_index } = slope_theta .* ( 0:1:size( channel_segments{
                    frame_index, channel_index }, 1 ) - 1 );
155
156     end
157
158 end
159
160
161
162 %% Compute Instantaneous RPM
163
164 for channel_index = 1:1:size( p, 2 )
165     for frame_index = 1:1:size( frame_set_indices, 1 )
166         rpm_estimate{ frame_index, channel_index } = 60 * ( 1 / ( numel( channel_segments{ frame_index,
                channel_index } ) ./ fs ) );
167     end
168 end
169
170
171 figure( 'Name', 'Shaft Speed' ); ...
172
173     yyaxis left; ...
174         plot( [ rpm_estimate{ :, 1 } ] ./ 60 );  grid on;
```

```
175         ylabel( 'Cycles Per Second [CPS]' );
176         ylim( [ 0 65 ] );
177 
178     yyaxis right; ...
179         plot( [ rpm_estimate{ :, 1 } ] );  grid on;
180         ylabel( 'Rotations Per Minute [RPM]' );
181 
182     xlabel( 'Revolution [WU]' );
183 
184     axis( [ -50 number_of_revolutions+50  0 3.75e3 ] );
185     shg;
186 
187 
188 
189 %% Compute Fourier Coefficients
190 
191 An = nan( size( frame_set_indices, 1 ), 11 );  Bn = An;
192 
193 TRACKING_ORDER = 2;
194 % TRACKING_ORDER = 4;
195 
196 for channel_index = 1:1:size( p, 2 )
197 
198     for frame_index = 1:1:size( frame_set_indices, 1 )
199 
200         M = numel( channel_segments{ frame_index, channel_index } );
201 
202         An( frame_index, channel_index ) = ...
203             2/M * channel_segments{ frame_index, channel_index }.' * cos( TRACKING_ORDER/2 .* theta{
                    frame_index, channel_index } ).';
204 
205         Bn( frame_index, channel_index ) = ...
206             2/M * channel_segments{ frame_index, channel_index }.' * sin( TRACKING_ORDER/2 .* theta{
                    frame_index, channel_index } ).';
207 
208     end
209 
210 end
211 
212 
213 
214 %% Plot Fourier Coefficients An and Bn and the Overall Sound Pressure Level for Channel 1
215 
216 if ( TRACKING_ORDER == 2 )
217 
218     figure( 'Name', 'Second-order Tracking and Associated Sound Pressure Level for Channel 1' ); ...
219 
220         subplot( 2, 1, 1 ); ...
221             plot( An( :, 1 ), 'Color', 'b' );  hold on;
222             plot( Bn( :, 1 ), 'Color', 'r' );  grid on;
223                 legend( '2x SR (cosine, $A_n$)', '2x SR (sine, $B_n$)', 'Location', 'South', '
                        Interpreter', 'Latex' );
224             xlabel( 'Revolution [WU]' );  ylabel( 'Fourier Coefficient' );
225             grid on;  axis( [ 1 number_of_revolutions  -0.3 0.4 ] );
226 
227         subplot( 2, 1, 2 ); ...
228             plot( 10*log10( ( An( :, 1 ).^2 + Bn( :, 1 ).^2 ) ./ 2e-6^2 ), 'Color', 'k' );  grid on;
229             xlabel( 'Revolution [WU]' );  ylabel( 'Sound Pressure dB re:20e-6 Pa' );
230             axis( [ 1 number_of_revolutions  50 110 ] );
231 
232         shg;
233 
234 else
235 
236     figure( 'Name', 'Fourth-order Tracking and Associated Sound Pressure Level for Channel 1' ); ...
237 
238         subplot( 2, 1, 1 ); ...
239             plot( An( :, 1 ), 'Color', 'b' );  hold on;
240             plot( Bn( :, 1 ), 'Color', 'r' );  grid on;
241                 legend( '4x SR (cosine, $A_n$)', '4x SR (sine, $B_n$)', 'Location', 'South', '
                        Interpreter', 'Latex' );
242             xlabel( 'Revolution [WU]' );  ylabel( 'Fourier Coefficient' );
243             grid on;  axis( [ 1 number_of_revolutions  -1 1 ] );
244 
245         subplot( 2, 1, 2 ); ...
246             plot( 10*log10( ( An( :, 1 ).^2 + Bn( :, 1 ).^2 ) ./ 2e-6^2 ), 'Color', 'k' );  grid on;
247             xlabel( 'Revolution [WU]' );  ylabel( 'Sound Pressure dB re:20e-6 Pa' );
248             axis( [ 1 number_of_revolutions  50 120 ] );
249 
250         shg;
251 
252 
253 end
254 
255 
256 
257 %% Plot Fourier Coefficients An and Bn and the Overall Sound Pressure Level for all Channels
258 
259 ORDER = 3;  FRAMELEN = 25;
260 
261 FOURIER_OFFSET = 0.5;
262 MAGNITUDE_OFFSET = 25;
```

```
263
264
265 if ( TRACKING_ORDER == 2 )
266
267     figure( 'Name', 'Second-order Tracking and Associated Sound Pressure Level for all Channels' ); ...
268
269     CHANNEL_INDEX = 11;
270
271         h1 = subplot( 2, 2, 1 ); ...
272
273             for channel_index = 1:1:CHANNEL_INDEX
274
275                 if ( channel_index < 5 )
276                     h_above = plot( sgolayfilt( An( :, channel_index ), ORDER, FRAMELEN ) +
                            FOURIER_OFFSET, 'Color', 'b', 'LineStyle', '--' );  hold on;
277                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
278                 elseif ( channel_index == 5 )
279                     h_plane = plot( sgolayfilt( An( :, channel_index ), ORDER, FRAMELEN ), 'Color', 'b'
                            );  hold on;
280                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
281                 else
282                     h_below = plot( sgolayfilt( An( :, channel_index ), ORDER, FRAMELEN ) -
                            FOURIER_OFFSET, 'Color', 'b', 'LineStyle', '-.' );  hold on;
283                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
284                 end
285
286             end
287             %
288             legend( [ h_above h_plane h_below ], { '2x SR (cosine, $A_n$) - Positive Angles', '2x SR (
                    cosine, $A_n$) - Plane', '2x SR (cosine, $A_n$) - Negative Angles' }, 'Location', '
                    South', 'Interpreter', 'Latex' );
289             xlabel( 'Revolution [WU]' );  ylabel( 'Fourier Coefficient' );
290             grid on;  axis( [ 1 number_of_revolutions  -2 1 ] );
291
292         % fprintf( 1, '\n\n' );
293
294         h2 = subplot( 2, 2, 2 ); ...
295
296             for channel_index = 1:1:CHANNEL_INDEX
297
298                 if ( channel_index < 5 )
299                     h_above = plot( sgolayfilt( Bn( :, channel_index ), ORDER, FRAMELEN ) +
                            FOURIER_OFFSET, 'Color', 'r', 'LineStyle', '--' );  hold on;
300                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
301                 elseif ( channel_index == 5 )
302                     h_plane = plot( sgolayfilt( Bn( :, channel_index ), ORDER, FRAMELEN ), 'Color', 'r'
                            );  hold on;
303                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
304                 else
305                     h_below = plot( sgolayfilt( Bn( :, channel_index ), ORDER, FRAMELEN ) -
                            FOURIER_OFFSET, 'Color', 'r', 'LineStyle', '-.' );  hold on;
306                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
307                 end
308
309             end
310             %
311             legend( [ h_above h_plane h_below ], { '2x SR (sine, $B_n$) - Positive Angles', '2x SR (
                    sine, $B_n$) - Plane', '2x SR (sine, $B_n$) - Negative Angles' }, 'Location', 'South',
                    'Interpreter', 'Latex' );
312             xlabel( 'Revolution [WU]' );  ylabel( 'Fourier Coefficient' );
313             grid on;  axis( [ 1 number_of_revolutions  -2 1 ] );
314
315         % fprintf( 1, '\n\n' );
316
317         h3 = subplot( 2, 2, [ 3 4 ] ); ...
318
319             for channel_index = 1:1:CHANNEL_INDEX
320
321                 if ( channel_index < 5 )
322                     h_above = plot( sgolayfilt( 10*log10( ( An( :, channel_index ).^2 + Bn( :,
                            channel_index ).^2 ) ./ 2e-6^2 ) + MAGNITUDE_OFFSET, ORDER, FRAMELEN ), 'Color'
                            , 'k', 'LineStyle', '--' );  hold on;
323                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
324                 elseif ( channel_index == 5 )
325                     h_plane = plot( sgolayfilt( 10*log10( ( An( :, channel_index ).^2 + Bn( :,
                            channel_index ).^2 ) ./ 2e-6^2 ), ORDER, FRAMELEN ), 'Color', 'k' );  hold on;
326                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
327                 else
328                     h_below = plot( sgolayfilt( 10*log10( ( An( :, channel_index ).^2 + Bn( :,
                            channel_index ).^2 ) ./ 2e-6^2 ) - MAGNITUDE_OFFSET, ORDER, FRAMELEN ), 'Color'
                            , 'k', 'LineStyle', '-.' );  hold on;
329                         % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index
                                ) );
330                 end
```

```
331
332             end
333             %
334             legend( [ h_above h_plane h_below ], { 'Magnitude Response - Positive Angles', 'Magnitude Response - Plane', 'Magnitude Response - Negative Angles' }, 'Location', 'South', 'Interpreter', 'Latex' );
335             xlabel( 'Revolution [WU]' );  ylabel( 'Sound Pressure Level [dB re: 20^-$\mu$ Pa]' );
336             grid on;  axis( [ 1 number_of_revolutions  20 140 ] );
337
338         shg;
339
340 else
341
342     FOURIER_OFFSET = 1;
343
344     figure( 'Name', 'Fourth-order Tracking and Associated Sound Pressure Level for all Channels' ); ...
345
346     CHANNEL_INDEX = 11;
347
348         h1 = subplot( 2, 2, 1 ); ...
349
350             for channel_index = 1:1:CHANNEL_INDEX
351
352                 if ( channel_index < 5 )
353                     h_above = plot( sgolayfilt( An( :, channel_index ), ORDER, FRAMELEN ) + FOURIER_OFFSET, 'Color', 'b', 'LineStyle', '--' );  hold on;
354                     % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index ) );
355                 elseif ( channel_index == 5 )
356                     h_plane = plot( sgolayfilt( An( :, channel_index ), ORDER, FRAMELEN ), 'Color', 'b' );  hold on;
357                     % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index ) );
358                 else
359                     h_below = plot( sgolayfilt( An( :, channel_index ), ORDER, FRAMELEN ) - FOURIER_OFFSET, 'Color', 'b', 'LineStyle', '-.' );  hold on;
360                     % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index ) );
361                 end
362
363             end
364             %
365             legend( [ h_above h_plane h_below ], { '4x SR (cosine, $A_n$) - Positive Angles', '4x SR (cosine, $A_n$) - Plane', '4x SR (cosine, $A_n$) - Negative Angles' }, 'Location', 'South', 'Interpreter', 'Latex' );
366             xlabel( 'Revolution [WU]' );  ylabel( 'Fourier Coefficient' );
367             grid on;  axis( [ 1 number_of_revolutions  -4 2 ] );
368
369         % fprintf( 1, '\n\n' );
370
371         h2 = subplot( 2, 2, 2 ); ...
372
373             for channel_index = 1:1:CHANNEL_INDEX
374
375                 if ( channel_index < 5 )
376                     h_above = plot( sgolayfilt( Bn( :, channel_index ), ORDER, FRAMELEN ) + FOURIER_OFFSET, 'Color', 'r', 'LineStyle', '--' );  hold on;
377                     % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index ) );
378                 elseif ( channel_index == 5 )
379                     h_plane = plot( sgolayfilt( Bn( :, channel_index ), ORDER, FRAMELEN ), 'Color', 'r' );  hold on;
380                     % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index ) );
381                 else
382                     h_below = plot( sgolayfilt( Bn( :, channel_index ), ORDER, FRAMELEN ) - FOURIER_OFFSET, 'Color', 'r', 'LineStyle', '-.' );  hold on;
383                     % fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index ) );
384                 end
385
386             end
387             %
388             legend( [ h_above h_plane h_below ], { '4x SR (sine, $B_n$) - Positive Angles', '4x SR (sine, $B_n$) - Plane', '4x SR (sine, $B_n$) - Negative Angles' }, 'Location', 'South', 'Interpreter', 'Latex' );
389             xlabel( 'Revolution [WU]' );  ylabel( 'Fourier Coefficient' );
390             grid on;  axis( [ 1 number_of_revolutions  -4 2 ] );
391
392         fprintf( 1, '\n\n' );
393
394         h3 = subplot( 2, 2, [ 3 4 ] ); ...
395
396             for channel_index = 1:1:CHANNEL_INDEX
397
398                 if ( channel_index < 5 )
399                     h_above = plot( sgolayfilt( 10*log10( ( An( :, channel_index ).^2 + Bn( :, channel_index ).^2 ) ./ 2e-6^2 ) + MAGNITUDE_OFFSET, ORDER, FRAMELEN ), 'Color', 'k', 'LineStyle', '--' );  hold on;
400                     fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index ) );
401                 elseif ( channel_index == 5 )
```

```
402                     h_plane = plot( sgolayfilt( 10*log10( ( An( :, channel_index ).^2 + Bn( :,
                            channel_index ).^2 ) ./ 2e-6^2 ), ORDER, FRAMELEN ), 'Color', 'k' );  hold on;
403                     fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index )
                            );
404                 else
405                     h_below = plot( sgolayfilt( 10*log10( ( An( :, channel_index ).^2 + Bn( :,
                            channel_index ).^2 ) ./ 2e-6^2 ) - MAGNITUDE_OFFSET, ORDER, FRAMELEN ), 'Color'
                            , 'k', 'LineStyle', '-.' );  hold on;
406                     fprintf( 1, '\n%d - %3.1f', channel_index, mic_angles_degrees( channel_index )
                            );
407                 end
408
409             end
410             %
411             legend( [ h_above h_plane h_below ], { 'Magnitude Response - Positive Angles', 'Magnitude
                    Response - Plane', 'Magnitude Response - Negative Angles' }, 'Location', 'South', '
                    Interpreter', 'Latex' );
412             xlabel( 'Revolution [WU]' );  ylabel( 'Sound Pressure Level [dB re: 20^$\mu$ Pa]' );
413             grid on;  axis( [ 1 number_of_revolutions  20 150 ] );
414
415         shg;
416
417 end
418
419
420
421 %% Clean-up
422
423 if ( ~isempty( findobj( 'Type', 'figure' ) ) )
424     monitors = get( 0, 'MonitorPositions' );
425         if ( size( monitors, 1 ) == 1 )
426             autoArrangeFigures( 3, 4, 1 );
427         elseif ( 1 < size( monitors, 1 ) )
428             autoArrangeFigures( 2, 2, 2 );
429         end
430 end
431
432 fprintf( 1, '\n\n\n*** Processing Complete ***\n\n\n' );
433
434
435
436 %% Reference(s)
437
438 % https://www.mathworks.com/help/signal/ref/ordertrack.html#bvaw5ra-1-x
439
440 % https://www.mathworks.com/matlabcentral/answers/1439884-evaluate-rising-edge-sample-of-a-signal
```